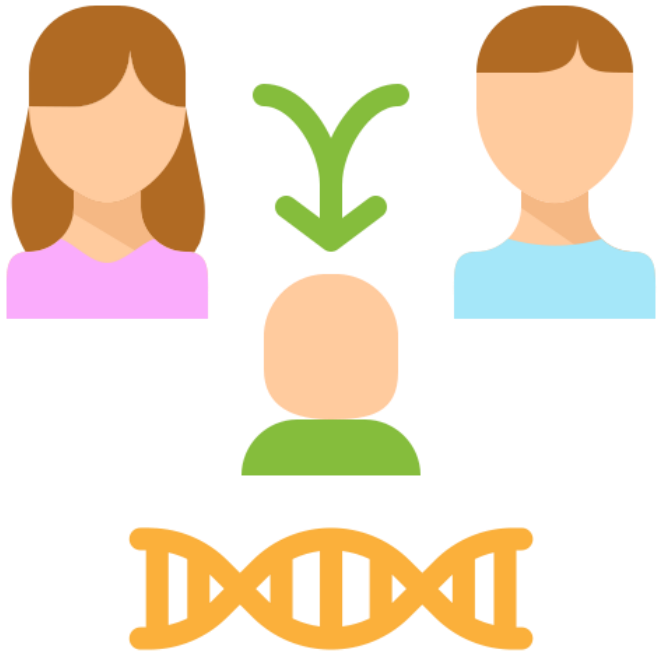
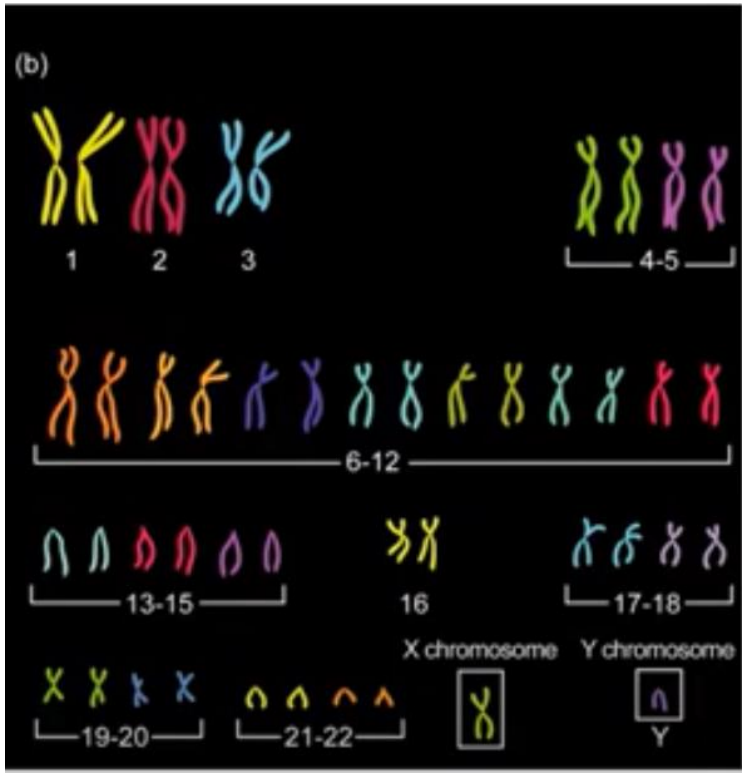
Genetics and Heredity - 3

Lecture 08

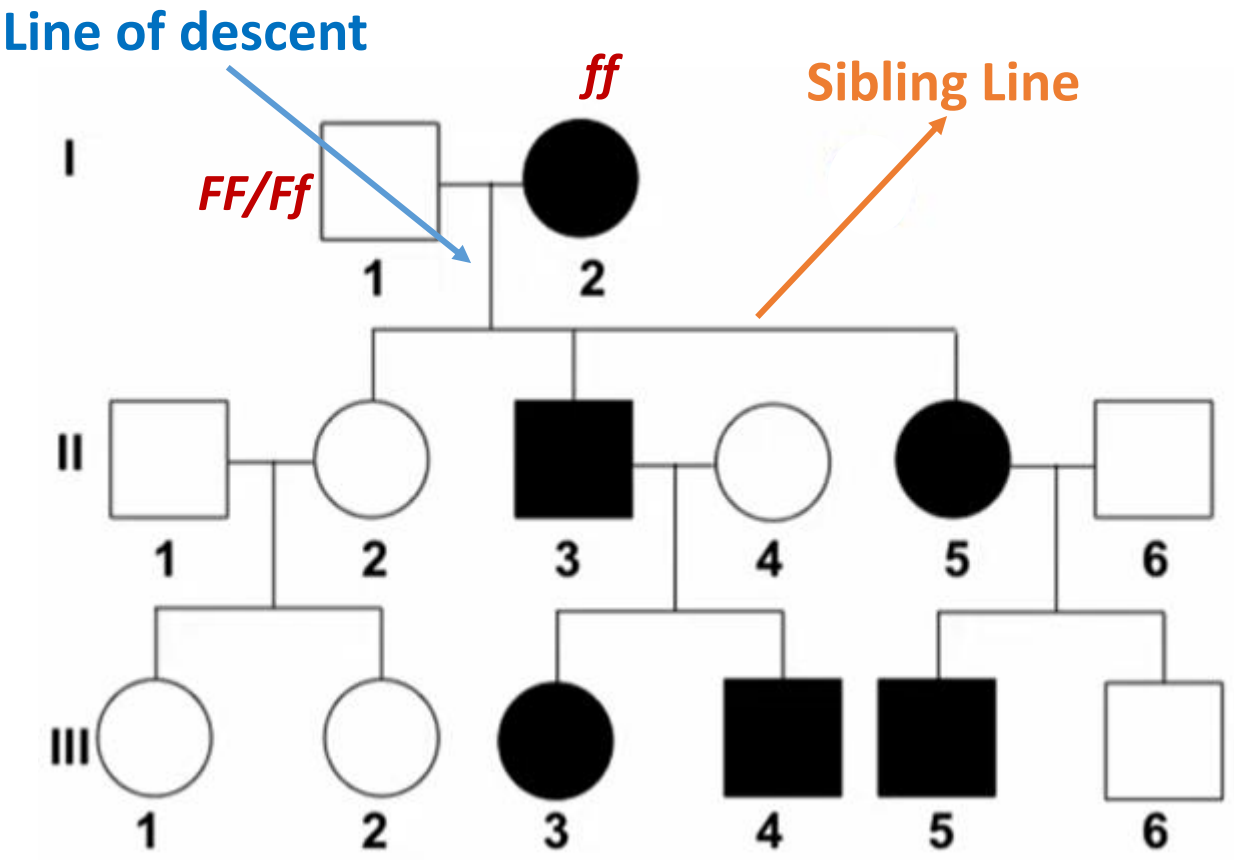


Chromosomes – some facts

- Humans have 46 chromosomes – 23 pairs
- Out of the 23 pairs of chromosomes, one pair contains the sex chromosomes.
- The non-sex chromosomes are referred to as autosomes
  - Traits linked to autosomes are called autosomal traits
- The sex-chromosomes are referred as ‘X’ and ‘Y’ chromosomes
  - Besides sex information, there are almost 900 – 1400 genes in the X chromosome, and about 70 – 140 in the Y chromosome
  - Traits and genes linked to the X chromosome are referred to as “X-linked.”
  - Traits and genes linked to the Y chromosome are referred to as “Y-linked.”

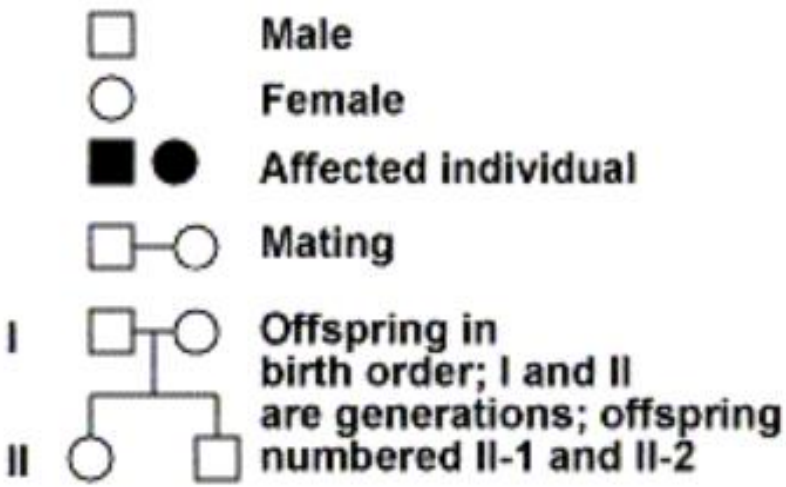


Pedigree analysis









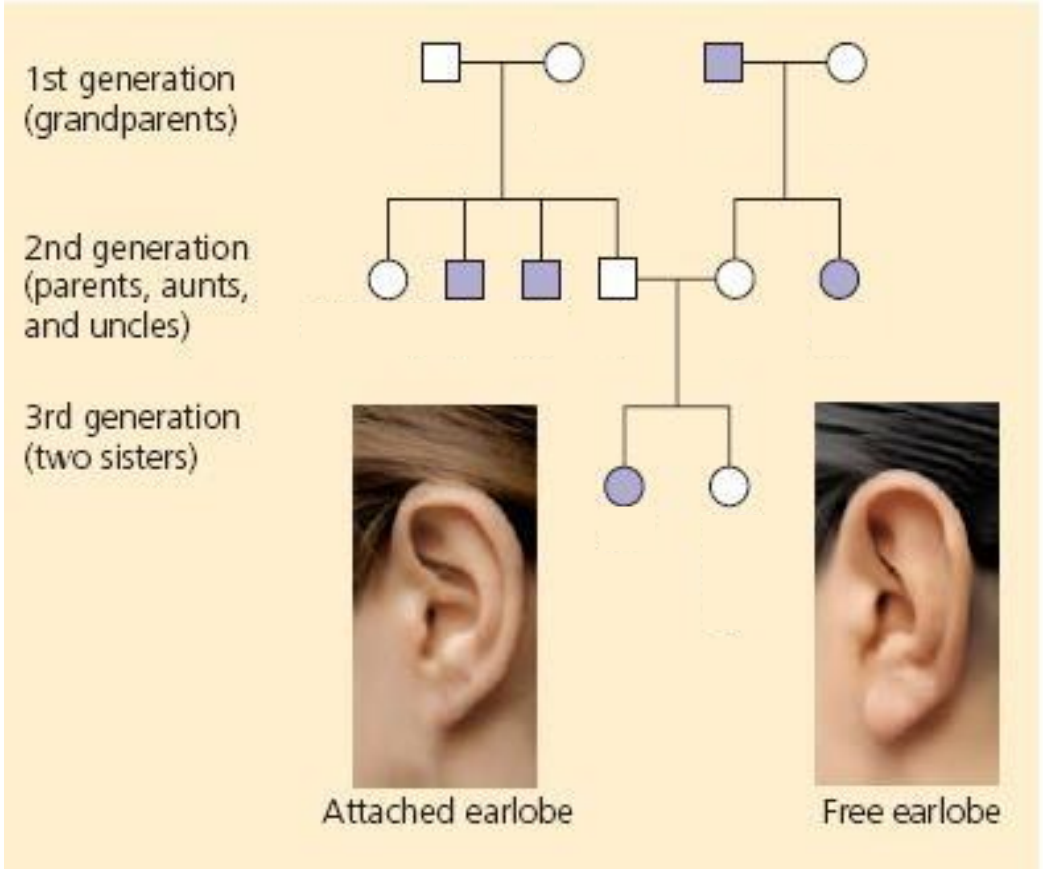
Pedigree analysis in genetics is the study of how traits or disorders are passed down through families. The genotypes are worked out as completely as possible and used to make further predictions.

Autosomal traits

Attached earlobes. recessive trait.

f for the recessive allele and F for the dominant allele.

- According to Mendelian inheritance, only homozygous (ff) will show phenotype of attached earlobe.
- First generation must be heterozygous (Ff) as second generation shows attached earlobes (ff).
- Medelian law is helpful for pedigree analysis.



Autosomal traits

Widow's peak : Occurrence of a pointed contour of the hairline on the forehead.

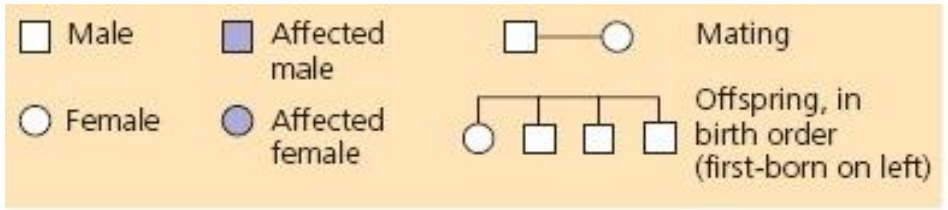
This trait, is due to a dominant allele, W.

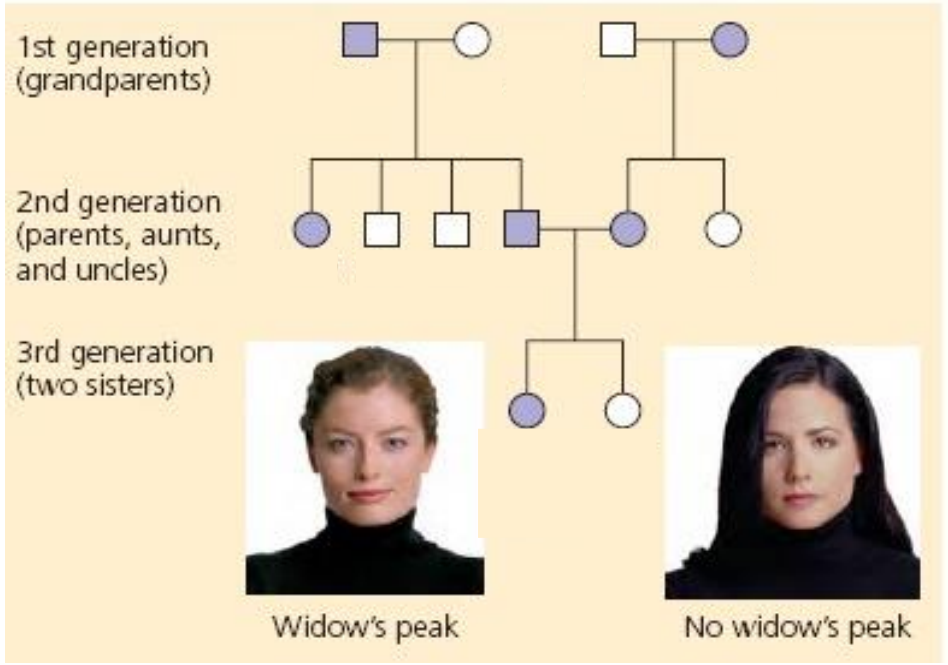
So, all individuals who lack a widow's peak must be homozygous recessive (ww).

The two grandparents with widow's peaks must have the Ww genotype, since some of their offspring are homozygous recessive.

Second generation who do have widow's peaks must also be heterozygous, because they are the products of Ww x ww matings.

The third generation consists of two sisters one has a widow's peak, could be either homozygous (WW) or heterozygous (Ww).





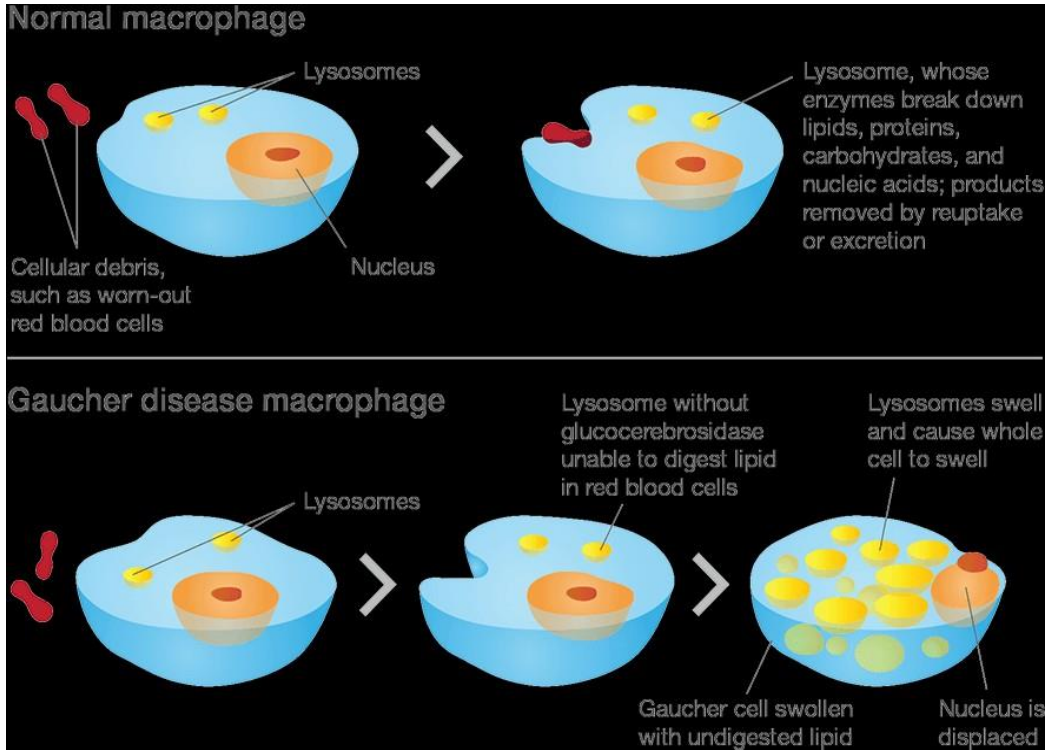
Gaucher Disease

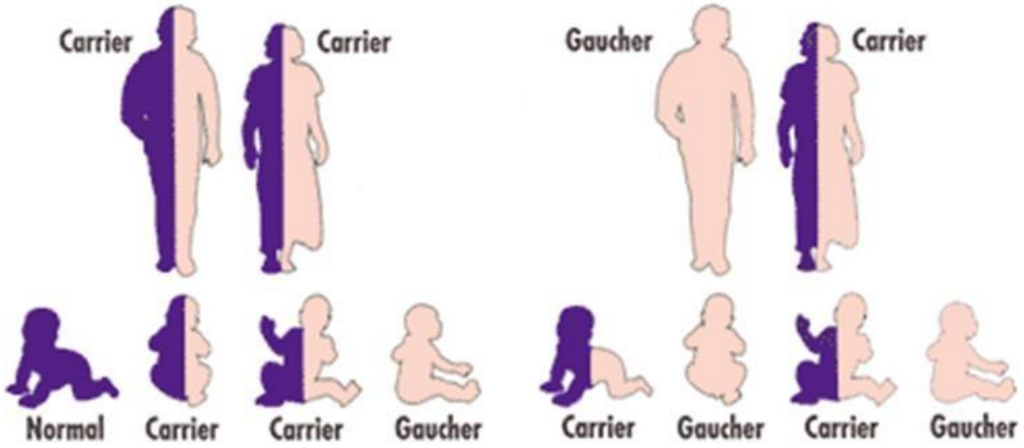
Gaucher Disease is a rare, genetic disease. It causes lipid-storage disorder (lipids accumulate in spleen, liver, bone marrow) due to deficiency of the enzyme glucocerebrosidase.

Gaucher disease is an autosomal recessive disorder (g).

When one Gaucher gene in the pair is abnormal (Gg), the child is an asymptomatic carrier who can pass that abnormal gene to his or her children.

When both genes in the pair are defective, a person will have Gaucher disease (gg).



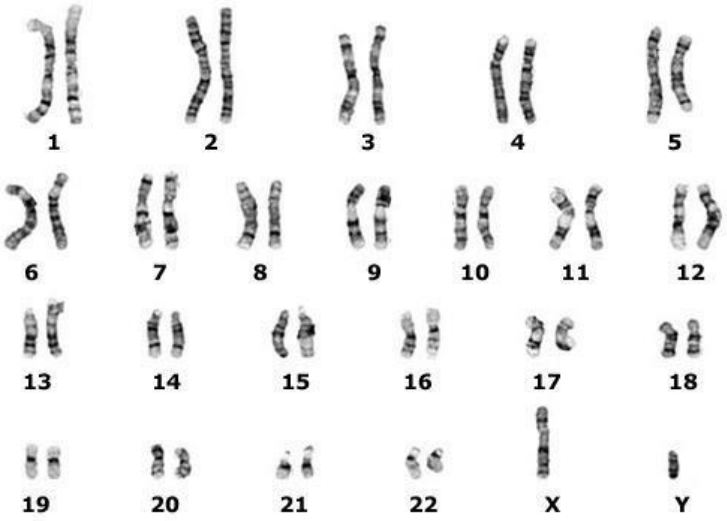


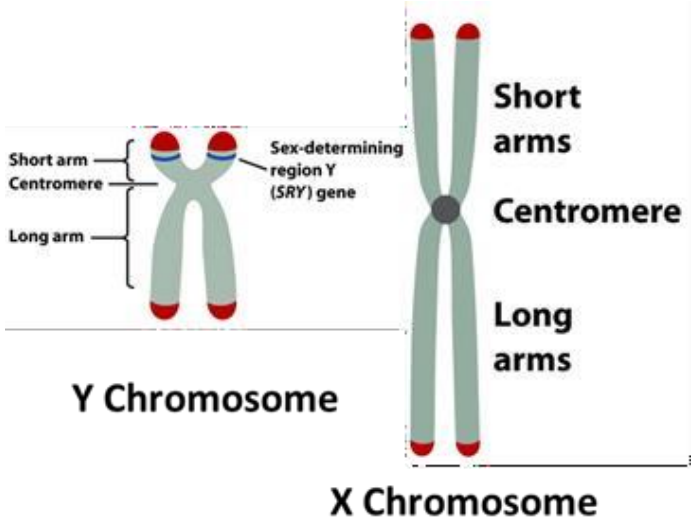
Sex Determination in Human

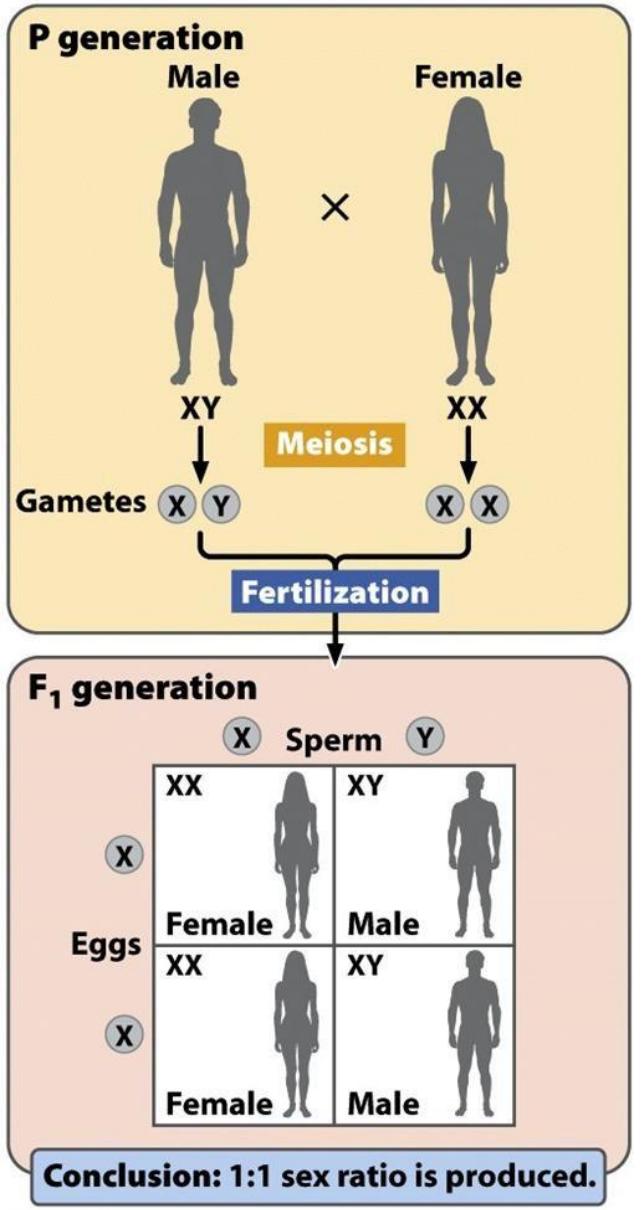
In humans, each cell normally contains 23 pairs of chromosomes, for a total of 46. Twenty-two of these pairs, called autosomes, look the same in both males and females.

The 23rd pair, the sex chromosomes, differ between males and females.

Females have two copies of the X chromosome, while males have one X and one Y chromosome.







Color Blindness: Sex-Linked Inheritance

Color Blindness is a sex-linked recessive trait found on the X chromosome.

X^+X^+ = Normal Female

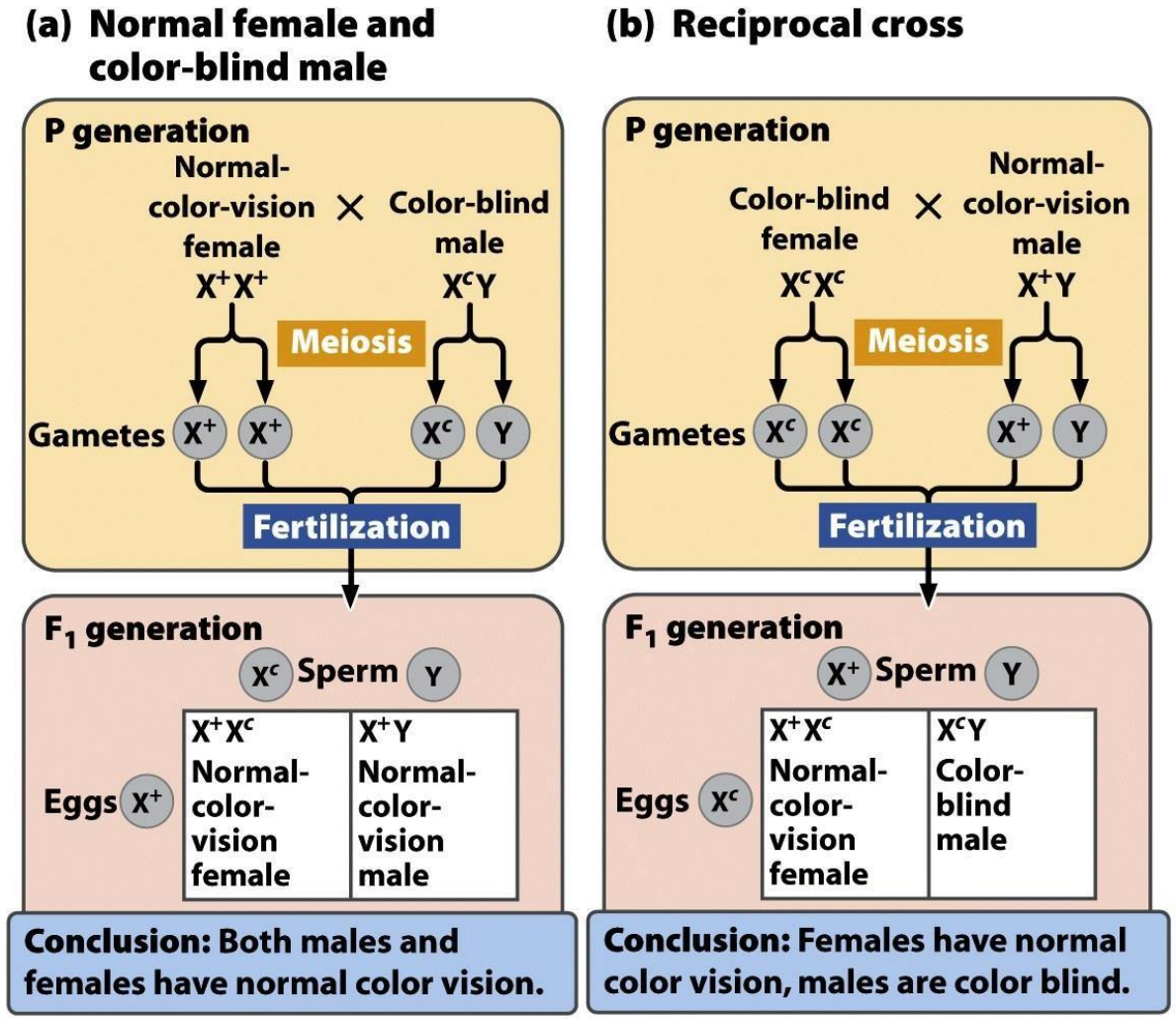
X^+X^c = Female Carrier

X^cX^c = Colour blind female

X^+Y = Normal Male

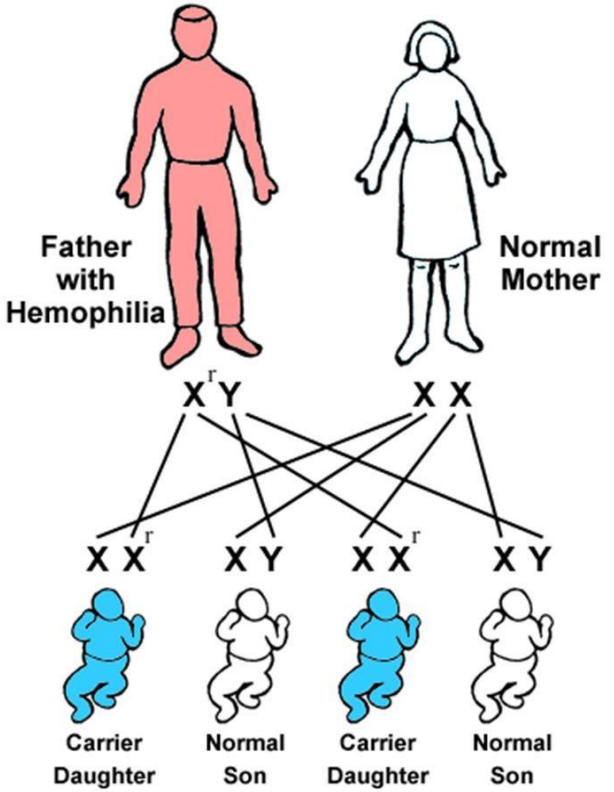
X^cY = Colour blind Male

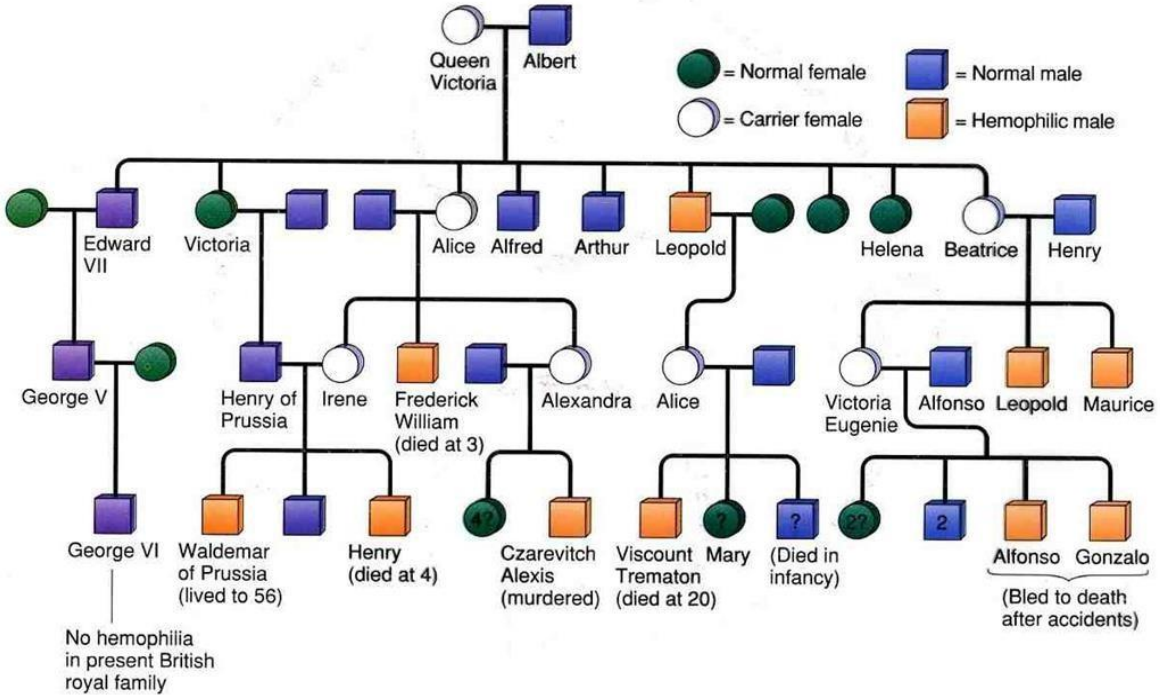
Males are more likely to be color blind then females as they only have one X chromosome.



Hemophilia:Sex-Linked Inheritance

- A blood disorder where the blood does not clot properly.
- A minor cut can cause serious injury.
- Internal bleeding and deep cuts can be fatal for hemophiliacs.
- Genetic, lack of one of the clotting factors.
- The gene for hemophilia is found on the X chromosome
- It is a sex-linked recessive disorder.

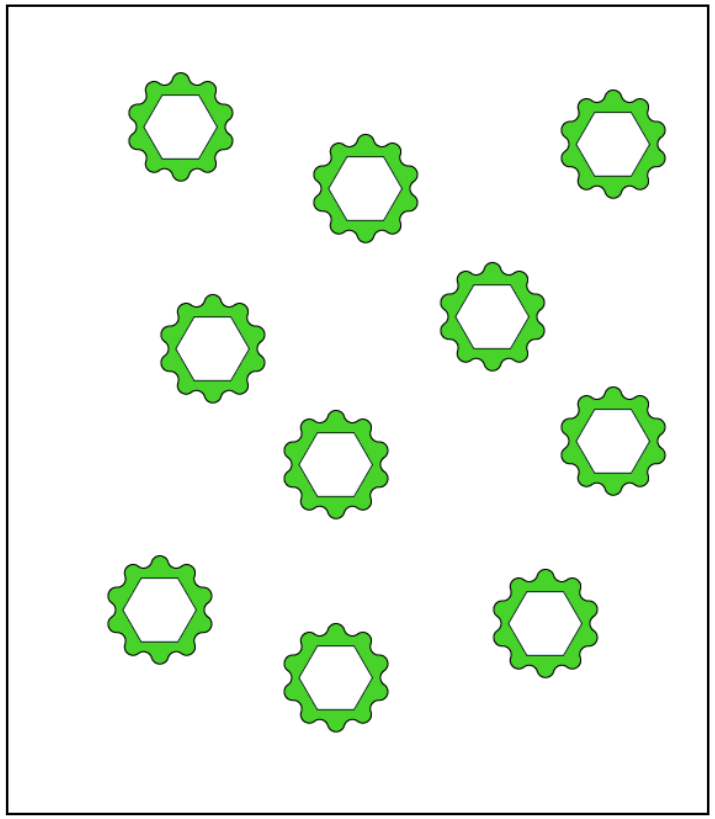




Post-Mendel search for heredity

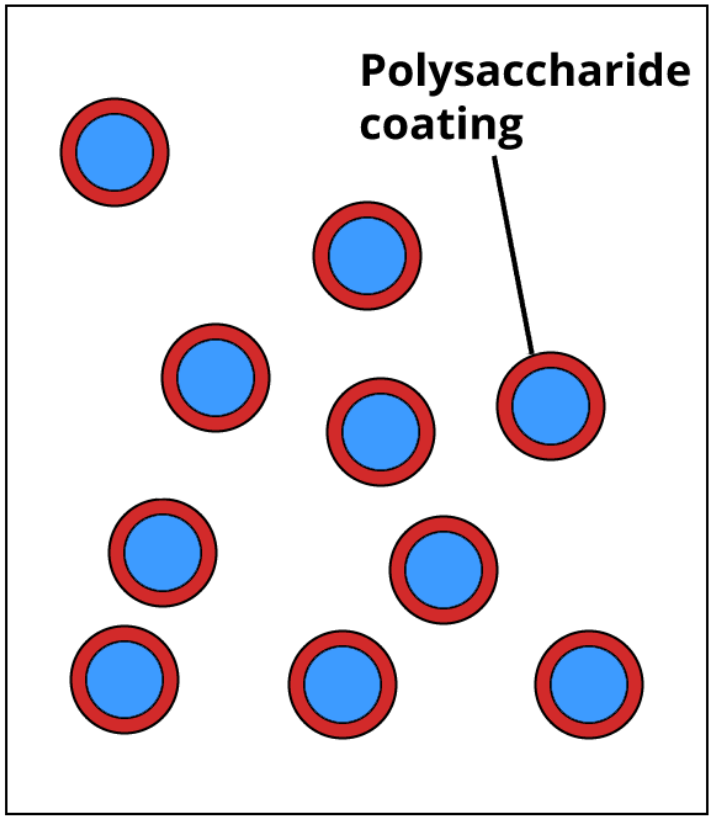
Griffith's Experiment (1928)

Streptococcus Pneumoniae



R strain (rough)

Non-Virulent

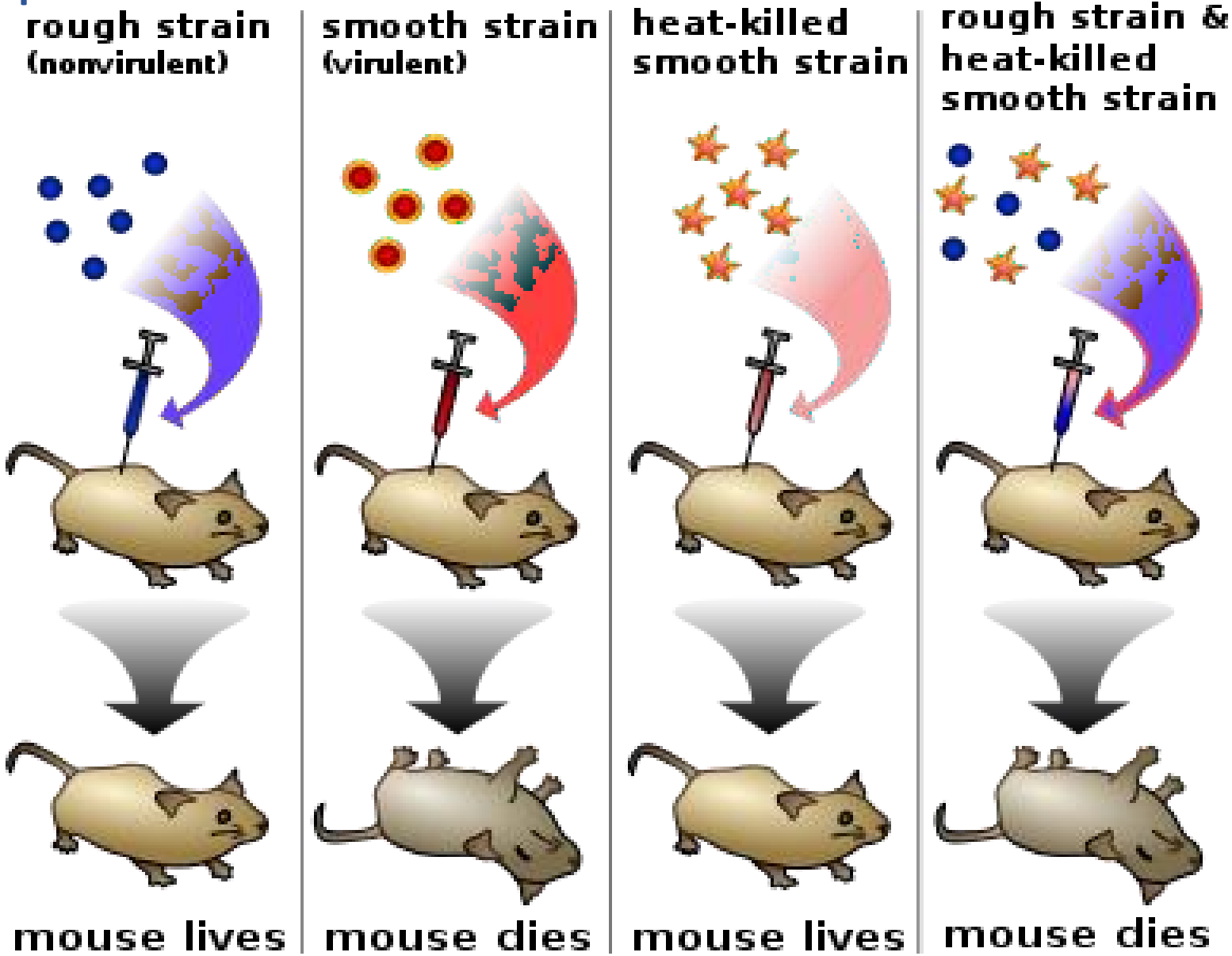


S strain (smooth)

Virulent

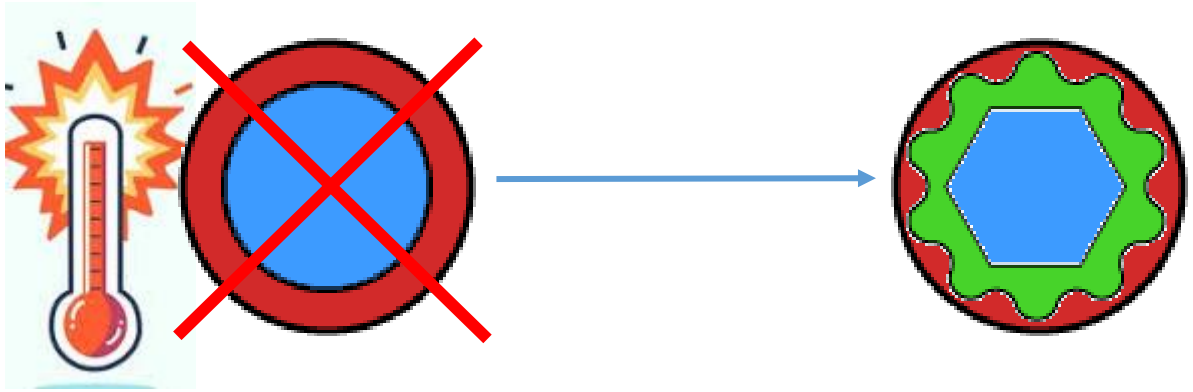


In 1928, British bacteriologist Frederick Griffith conducted a series of experiments using Streptococcus pneumoniae bacteria and mice.



https://en.wikipedia.org/wiki/Avery%E2%80%93MacLeod%E2%80%93McCarty_experiment

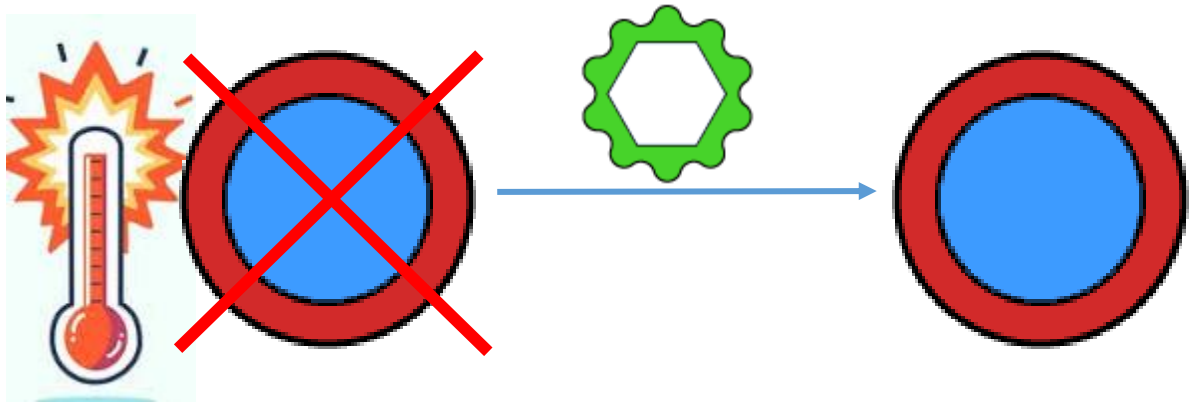
In 1928, British bacteriologist Frederick Griffith conducted a series of experiments using Streptococcus pneumoniae bacteria and mice.



The heat killed S-strain

Can transform the non-virulent R-strain to virulent S-strain

Griffith concluded that the R-strain bacteria must have taken up what he called a "transforming principle" from the heat-killed S bacteria, which allowed them to "transform" into smooth-coated bacteria and become virulent.



The heat killed S-strain

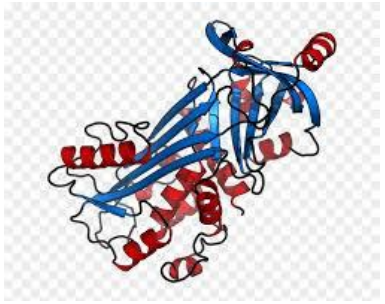
Can transform the non-virulent R-strain to virulent S-strain

- Transforming principle makes a harmless bacteria deadly!
- What could that transforming principle be?
- What kind of biomolecules or substance changes the characteristics of an organism?

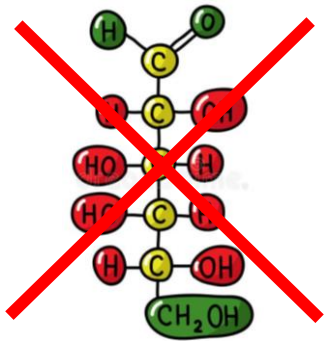
Avery-MacLeod-McCarty Experiment (1944)

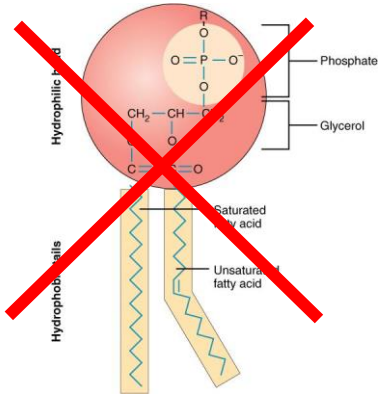
- The Avery, McLeod and McCarty experiment expanded upon Griffith's ideas and identified the molecule of inheritance.
- They isolated the mysterious material and identified it using extensive set of experiments.
- They examined the solubility, stability and compared the results from "mystery molecule" to the properties of known samples of biomolecules.



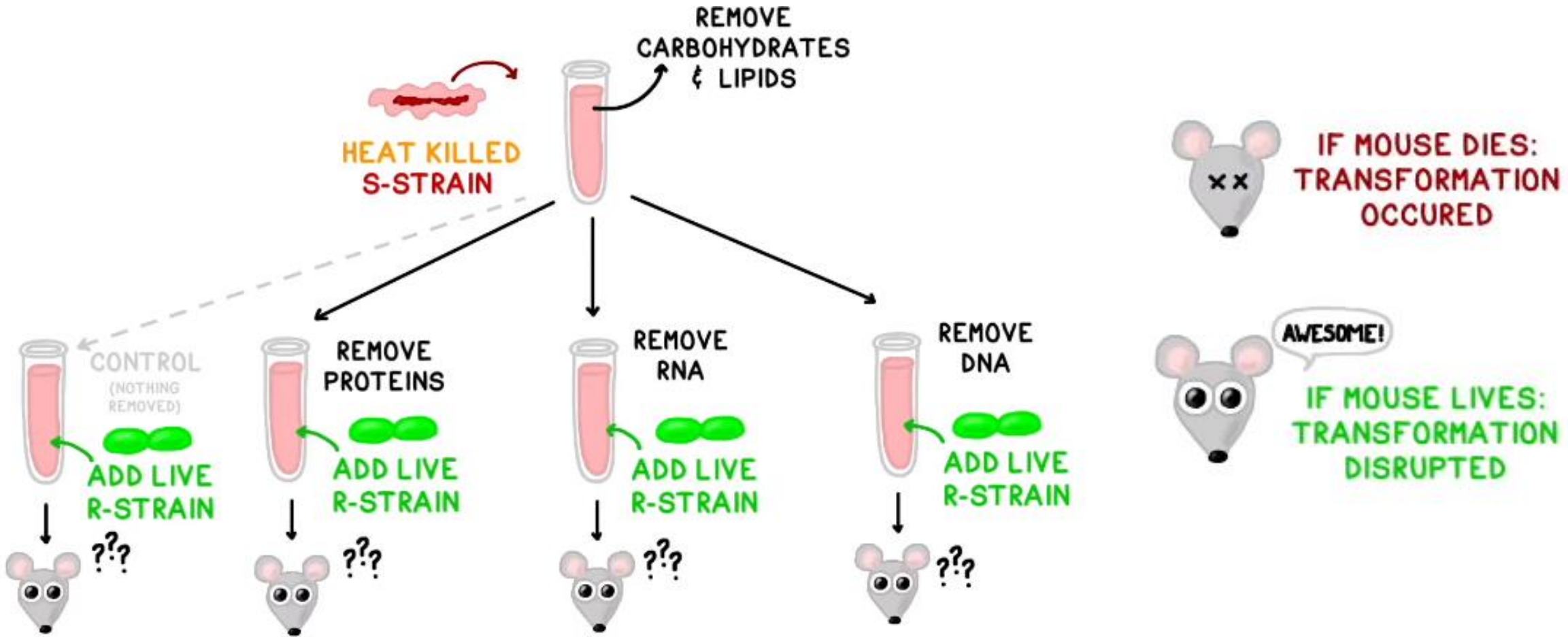








Avery-MacLeod-McCarty Experiment



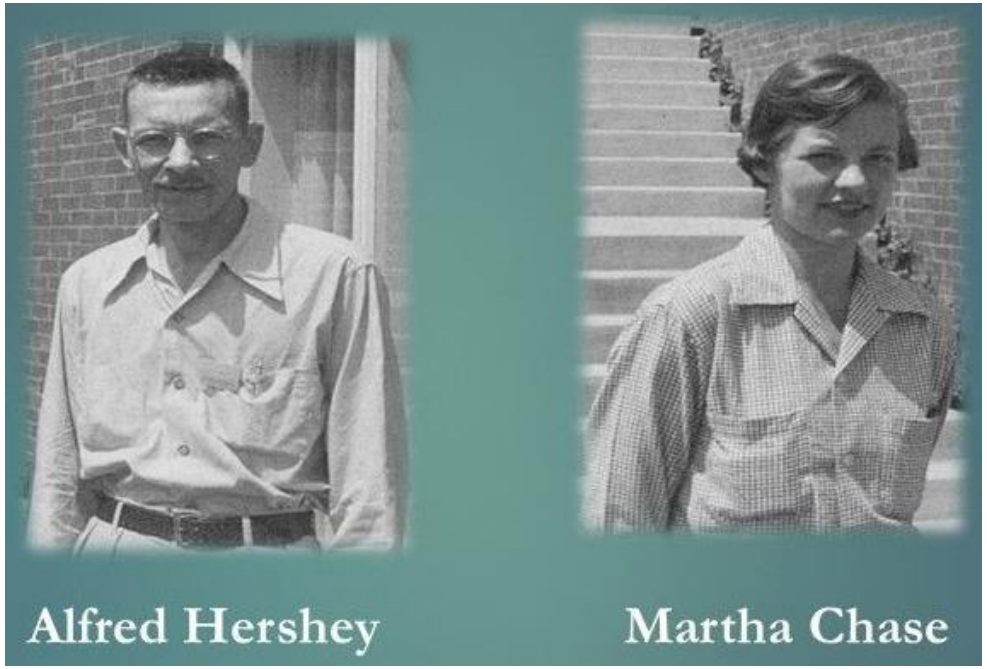
Avery-MacLeod-McCarty Experiment

COMPONENT	ENZYMES	RESULT	CONCLUSION
CONTROL	NO COMPONENTS DESTROYED	 MOUSE DIED: VIRULENT STRAIN	N / A
PROTEINS	PROSTEASES: TRYPSIN & CHYMOTRYPSIN	 MOUSE DIED: VIRULENT STRAIN	PROTEINS ARE NOT THE T.P.
NUCLEIC ACID: RNA	RNASES: RIBONUCLEASE	 MOUSE DIED: VIRULENT STRAIN	RNA IS NOT THE T.P.
NUCLEIC ACID: DNA	DNASES: DEPOLYMERASES	 MOUSE LIVED: AVIRULENT STRAIN	DNA IS THE T.P.

T.P.: Transforming Principle

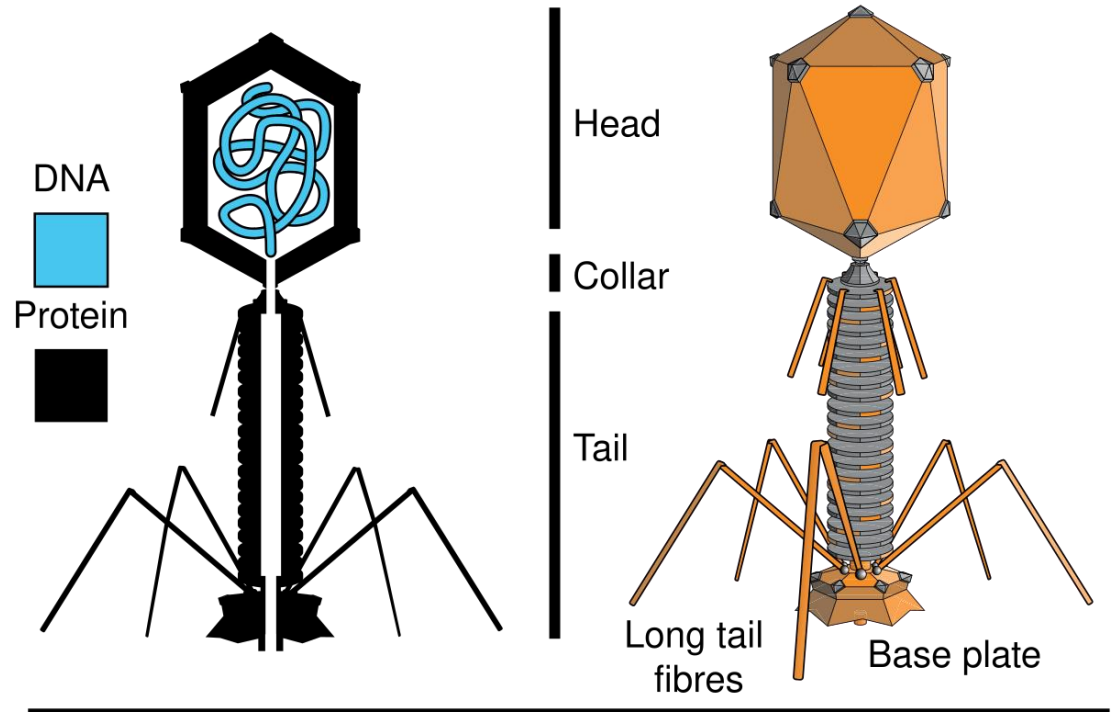
Hershey-Chase experiment (1952)

- Hershey-Chase experiment confirmed that DNA is the genetic material.
- They used T2 bacteriophage virus.
- Their experiment proved once and all that DNA is the genetic material



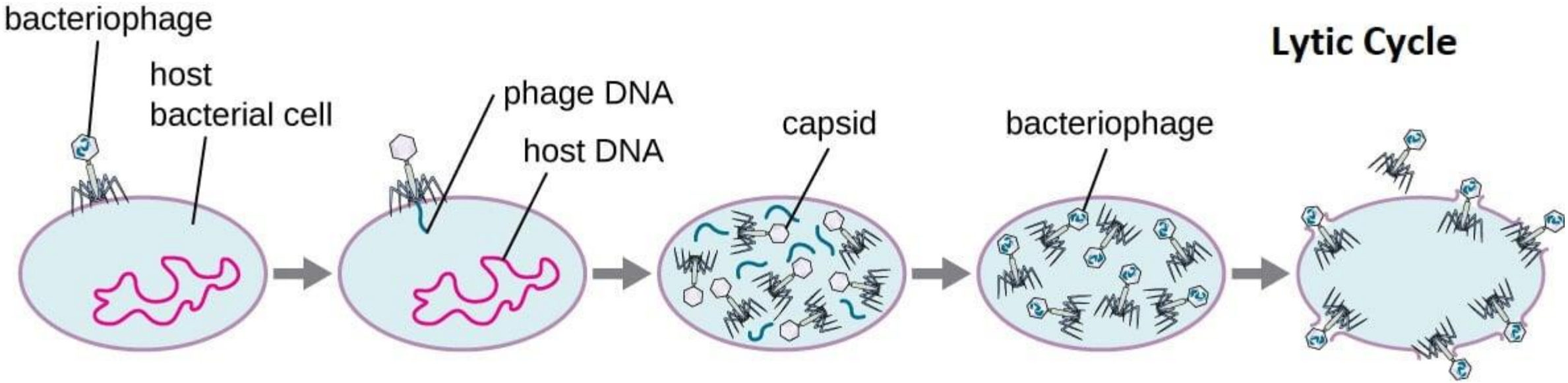
T2 bacteriophage

- They are viruses that infect bacteria.
- They consists of a head and a tail.
- Head contains the genetic material
- They reproduce quickly and contain only proteins and Nucleic acids



https://simple.wikipedia.org/wiki/T2_phage

Lytic Cycle



1. **Attachment** The phage attaches to the surface of the host.
2. **Penetration** The viral DNA enters the host cell.
3. **Biosynthesis** Phage DNA replicates and phage proteins are made.
4. **Maturation** New phage particles are assembled.
5. **Lysis** The cell lyses, releasing the newly made phages.

https://psu.pb.unizin.org/microb201/chapter/the-viral-life-cycle/

Hershey-Chase experiment – Principle

- DNA contains Phosphorous but no Sulphur
- Proteins contain Sulphur but no phosphorous
- Phosphorous and Sulphur have 31P and 32S as common isotopes, while 32P and 35S are their radioactive isotopes, respectively.



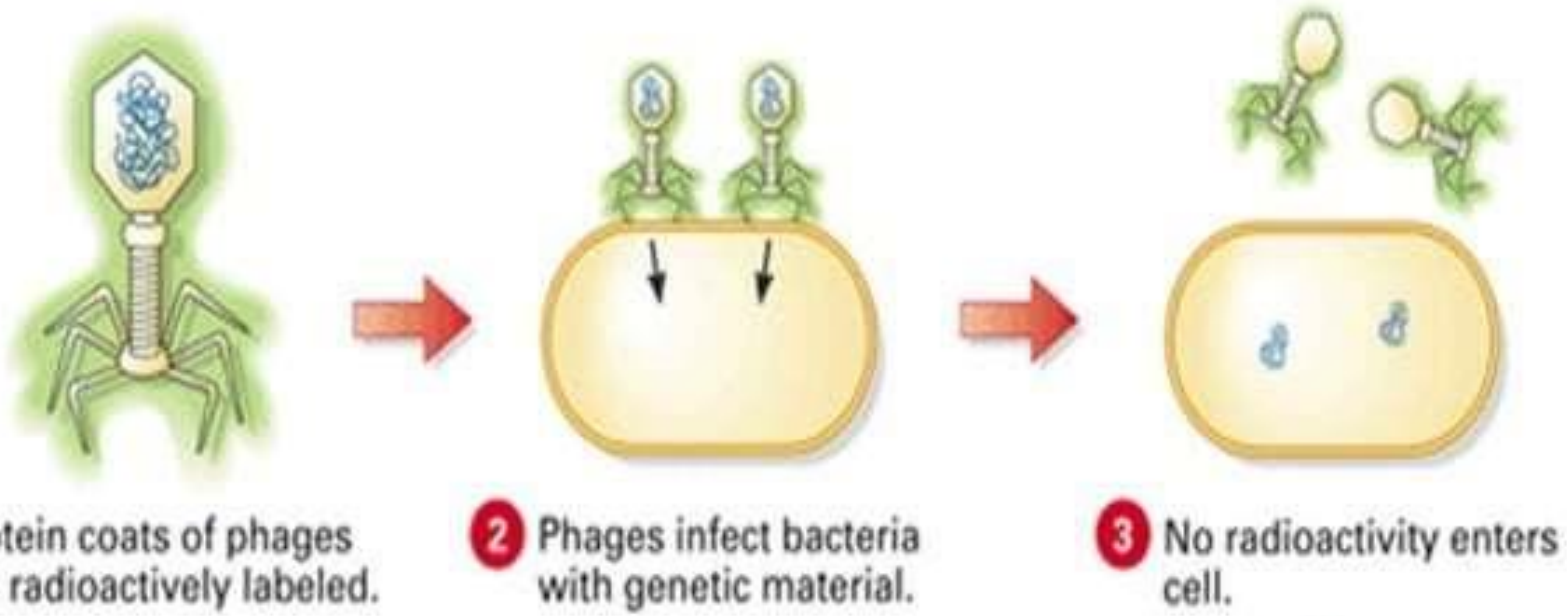


Hershey-Chase experiment

- Viral protein and DNA were tagged with different radioactive isotopes. Protein was tagged with radioactive Sulphur and DNA was tagged with radioactive phosphorous.
- Protein-laeled and DNA-labeled T2 phages were allowed to infect separate samples of non-radioactive E coli bacteria.
- Cultures were then agitated to shake loose phages that remained outside the bacterial cells.
- Mixtures were centrifuged forcing the heavier bacterial cells settle at the bottom of the tubes, while lighter viruses remained floating (supernatant).
- Radioactivity in the pellet and supernatant was measured and compared.







1 Protein coats of phages are radioactively labeled.

2 Phages infect bacteria with genetic material.

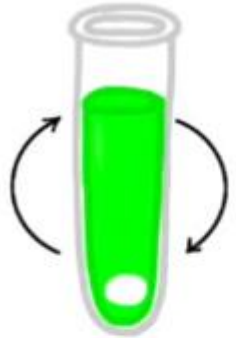
3 No radioactivity enters cell.



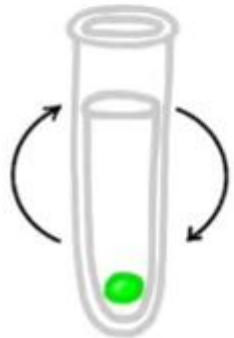
1 DNA of phages is radioactively labeled.

2 Phages infect bacteria with genetic material.

3 Radioactivity enters cell.



CENTRIFUGATION



Hershey-Chase experiment – Results

- In tubes with E Coli infected with protein-labeled T2, most of the radioactivity was in the supernatant with viruses.

- In tubes with E Coli infected with DNA-labeled T2, most of the radioactivity was in the pellet with bacterial cells.

This experiment settled once and for all that DNA is the genetic material!